THE FIRST ETALE COHOMOLOGY OF A FINITE CONSTANT GROUP ON A CLOSED SUBSCHEME OF A NORMAL AFFINE SCHEME WITH SEPARABLY CLOSED FUNCTION FIELD VANISHES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the translation of a cohomology class into a torsor and its description as a finite locally constant sheaf, the representation of the torsor by a finite etale scheme, and the closing trivialization through the existence of a section.

Editorial note. The label lemma-gabber-for-h1-absolutely-algebraically-closed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-gabber-for-h1-absolutely-algebraically-closed –start 14424 –end 14442 –local-id 3572 returned RESULT=OK (L3572 free, occupancy FREE, public interval clear). Candidate taken from operations/pool-cards/stacks-etale-cohomology-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The first cohomology with constant finite coefficients classifies torsors. Cohomology on Sites, the lemma identifying the first cohomology group with coefficients in a constant finite group with the set of torsors, and describing them as finite locally constant sheaves. Supplies the translation of a cohomology class into a torsor on the closed subscheme, which is a finite locally constant sheaf.
- Representability of a finite locally constant sheaf by a finite etale scheme. Same chapter, the lemma representing a finite locally constant sheaf by a finite etale morphism of schemes. Supplies the scheme V finite etale over Z representing the torsor, which is surjective over Z because a torsor is locally trivial.
- Sections of finite etale covers of a closed subscheme of a normal affine scheme. Same chapter, the lemma producing a section of a finite etale cover of a closed subscheme of a normal affine scheme with absolutely algebraically closed function field. Supplies the section of V -> Z, which makes the torsor trivial and hence the class zero.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 03RV: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9994–9994. Prerequisite pointer invoked at proof line(s) 14437; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09ZA: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 14390–14390. Prerequisite pointer invoked at proof line(s) 14440; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03AJ: sources/stacks-chow/tags.txt, source lines 4312–4312. Prerequisite pointer invoked at proof line(s) 14432; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a normal integral affine scheme with separably closed function field. Let $Z \subset X$ be a closed subscheme. For any finite abelian group M we have $H^1_{\acute{e}tale}(Z, \underline{M}) = 0$.*

Proof. By Cohomology on Sites, Lemma 03AJ an element of $H^1_{\acute{e}tale}(Z, \underline{M})$ corresponds to a $\underline{M}$-torsor $\mathcal{F}$ on $Z_{\acute{e}tale}$. Such a torsor is clearly a finite locally constant sheaf. Hence $\mathcal{F}$ is representable by a scheme V finite étale over Z, Lemma 03RV. Of course $V \to Z$ is surjective as a torsor is locally trivial. Since $V \to Z$ has a section by Lemma 09ZA we are done. □